Birla Institute of Technology and Science, Pilani

(Department of Management)

MPBA G512 –Time Series Analysis and Forecasting

Time Series Analysis of Inflation Rates across the Countries

Submitted by

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Problem Statement

In this project we consider the major global economies to gauge the effects of inflation across advanced and emerging economies. As ranked recently by Forbes, China and the United States are now recognized as being the world's two largest economies. Due to their size and reach into worldwide markets, volatility in their respective inflation rates holds far-reaching implications for the overall economy. In addition to these, we take the case of one of the fastest-growing large economies, India, to get a comparative insight into how inflation patterns have influenced an emerging economy. This choice enables us to study inflation behavior under different economic conditions.
This project investigates long-term inflation trends in the US,China, and India. The central goals are:

- To study the historical patterns of inflation in India.
- To model and forecast inflation rate using ARIMA.
- To put India's inflation pattern in perspective against other large economies.
- To test for stationarity and volatility in inflation series.

By way of this analysis, the project seeks to offer insightful information on India's economic path, situating it in the overall global inflationary context.

Background

To familiarize ourselves with the project, we conducted considerable background research in order to create a solid fundamental understanding of the economic as well as analytical elements at play. Preparation included:

A detailed examination of inflation and its macroeconomic significance, such as the numerous ways in which inflation is measured and how inflation tends to evolve over time within various economies and under varying economic conditions.

A quick guide to the basics of time series analysis, including key concepts like stationarity, autocorrelation, trend, seasonality, and volatility. These were key concepts to understand in order to properly preprocess the data and choose the model.

An exploration into ARIMA modeling — its theoretical foundations and practical implementation. We discussed the function of every component (AutoRegressive, Integrated, and Moving Average), differencing for stationarity, and methods for determining the appropriate parameters through AIC (Akaike Information Criterion) for model efficiency.

We also read applicable academic literature and real-world case studies in which ARIMA and other time series models were employed to predict economic indicators such as inflation. These sources gave us information on best practices and potential pitfalls, which informed our analytical approach.

This preparatory effort helped us to frame a well-designed project plan encompassing data acquisition, preprocessing, KPSS and ADF test stationarity checks, model estimation, validation, and ultimate forecasting.

Method

Data Source:

- CSV file inflation.csv containing inflation rates for countries from 1980–2022, downloaded from github.

Data Cleaning:

- Removed “no data” entries.
- Column names were cleaned to remove the “X” prefix.
- Converted the dataset from wide to long format using gather().
- Filtered country-specific data (India, China and US).

```
colnames(data) <- gsub("^X", "", colnames(data))
colnames(data)
data

year<-c(1980:2022)      #making a vector consisting of all years
year<-as.character(year)

inf<-data |>  gather(year,key = "Year",value="InflationRate")
inf1<-na.omit(inf) #omitting NA values

View(inf1)

inf_clean <- inf1 |>
  filter(InflationRate != "no data")

View(inf_clean)

names(inf_clean)<-c("region","year","inflation")
inf_clean$year<-as.integer(inf_clean$year)

India<-filter(inf_clean,region=="India")
India$inflation<-as.numeric(India$inflation)
India$year<-as.numeric(India$year)
```

Data Transformation:

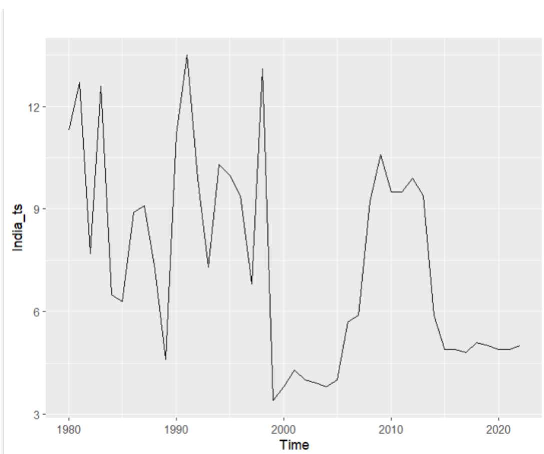
- Created individual time series objects for each country.
- Converted applicable variables (year and inflation rate) to numeric types.

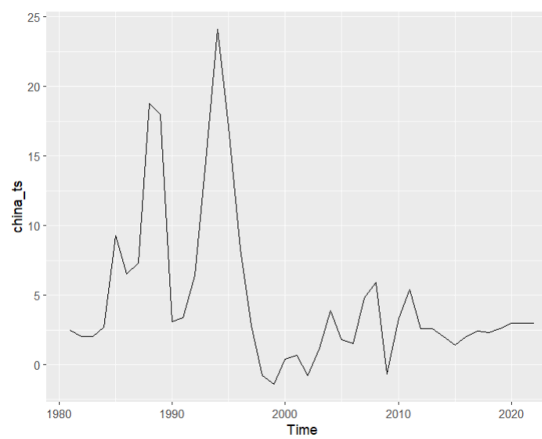
```
India<-filter(inf_clean,region=="India")
India$inflation<-as.numeric(India$inflation)
India$year<-as.numeric(India$year)
#log_ts<-log(India$inflation)
India_ts <- ts(India$inflation, start = min(India$year), end = max(India$year), frequency = 1)
(India_ts)
```

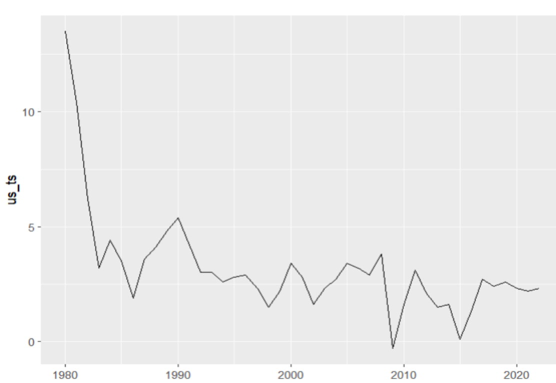
We have done the same for China and the US, and transformed to china_ts and us_ts.

Time Series Analysis:

- To start our examination, we graphed the inflation time series data for India and other chosen nations using the autoplot() function. These plots gave us a basic idea of the overall patterns, and any structural breaks apparent in the data over time.







In our analysis, we used multiple tests to assess stationarity:

KPSS Test:

- The KPSS test, which tests for stationarity under the null hypothesis, gave a test statistic of 0.4333, below the 5% critical value of 0.463. This means we failed to reject the null hypothesis, suggesting that the series is stationary.

```
> India_ts|> ur.kpss() |> summary()

#######################
# KPSS Unit Root Test #
#######################

Test is of type: mu with 3 lags.

Value of test-statistic is: 0.4333

Critical value for a significance level of:
                10pct  5pct 2.5pct  1pct
critical values 0.347 0.463  0.574 0.739

> |
```

```
> china_ts|> ur.kpss() |> summary()

#######################
# KPSS Unit Root Test #
#######################

Test is of type: mu with 3 lags.

Value of test-statistic is: 0.3447

Critical value for a significance level of:
                10pct  5pct 2.5pct  1pct
critical values 0.347 0.463  0.574 0.739
```

```
> US_ts|> ur.kpss() |> summary()

#######################
# KPSS Unit Root Test #
#######################

Test is of type: mu with 3 lags.

Value of test-statistic is: 0.7067

Critical value for a significance level of:
                10pct  5pct 2.5pct  1pct
critical values 0.347 0.463  0.574 0.739
```

ADF Test:

- The ADF test, with the null hypothesis of non-stationarity, gave a p-value of 0.06 (greater than 0.05). We failed to reject the null hypothesis, indicating non-stationarity.

```
> adf.test(India_ts)

        Augmented Dickey-Fuller Test

data:  India_ts
Dickey-Fuller = -2.2974, Lag order = 3, p-value = 0.4562
alternative hypothesis: stationary
```

```
> adf.test(US_ts)

        Augmented Dickey-Fuller Test

data:  US_ts
Dickey-Fuller = -3.3482, Lag order = 3, p-value = 0.07761
alternative hypothesis: stationary
```

```
> adf.test(china_ts)

        Augmented Dickey-Fuller Test

data:  china_ts
Dickey-Fuller = -3.5115, Lag order = 3, p-value = 0.05367
alternative hypothesis: stationary
```

ndiffs() Function:

- The ndiffs() function recommended 1 differencing, implying the series is non-stationary and requires one level of differencing.

For India_ts, china_ts, us_ts we are getting ndiffs() as 1

```
> nsdiffs(India_ts)
Error in nsdiffs(India_ts) : Non seasonal data
> ndiffs(India_ts )
[1] 1
>
```

```
> nsdiffs(china_ts)
Error in nsdiffs(china_ts) : Non seasonal data
> ndiffs(china_ts)
[1] 1
>
```

```
> nsdiffs(us_ts)
Error in nsdiffs(us_ts) : Non seasonal data
> ndiffs(us_ts)
[1] 1
>
```

This divergence between the KPSS and ADF tests is not uncommon and often signals that the data is trend stationary—stationary around a deterministic trend rather than a constant mean. Economic indicators such as inflation frequently exhibit this behavior.

Rather than treating these results as contradictory, we recognize that these tests provide complementary perspectives. While KPSS is sensitive to variance stability, ADF focuses on mean reversion. With the borderline outcome of the KPSS test and the strong suggestion from the ADF and ndiffs(), we chose to difference the series once to eliminate any trend component and satisfy the strict stationarity assumption needed for ARIMA modeling.

This conservative strategy prevents us from the under-differencing risks of model bias and inaccurate forecasting, while being consistent with best practice in macroeconomic time series analysis.

Modeling:

1. India – Inflation Time Series (India_ts)

Basic Modeling using auto.arima()

The inflation data was filtered to extract India-specific records and was subsequently transformed into a time series object. Initially, the auto.arima() function suggested an ARIMA(0,1,1) model.

Diagnostic Results for ARIMA(0,1,1)

The ARIMA(0,1,1) model did not meet the required diagnostic standards. This model appeared structurally appropriate because of AIC value(203.88), but it failed the Ljung-Box test, indicating that residual autocorrelations were still present.

```
> checkresiduals(fit1)

        Ljung-Box test

data:  Residuals from ARIMA(0,1,1)
Q* = 17.656, df = 8, p-value = 0.02396

Model df: 1.   Total lags used: 9
```

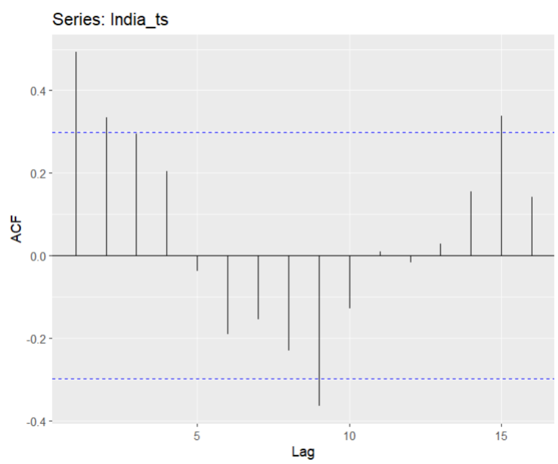
Here P = 0.02396 < alpha (0.05) So, the null hypothesis of Ljung box got rejected and concluded that the autocorrelation still exists between the residuals which is inappropriate for ARIMA modeling.

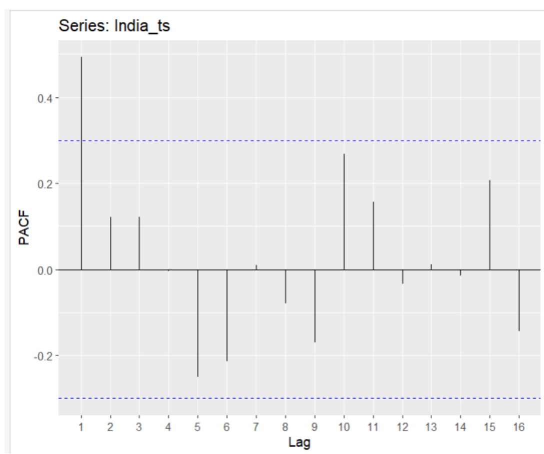
Exploration of ARIMA(2,1,1)

Based on the analysis of the ACF and PACF plots, the ARIMA(2,1,1) configuration was tested next. Furthermore, the AIC remained relatively high(207.42).

```
> fit3<- Arima(India_ts, order=c(2,1,1))
> fit3
Series: India_ts
ARIMA(2,1,1)

Coefficients:
          ar1      ar2      ma1
      -0.2600  -0.1782  -0.2265
s.e.   0.3866   0.2155   0.3748

sigma^2 = 7.222:  log likelihood = -99.71
AIC=207.42   AICc=208.5   BIC=214.37
>
```

The following are the ACF and PACF for India:





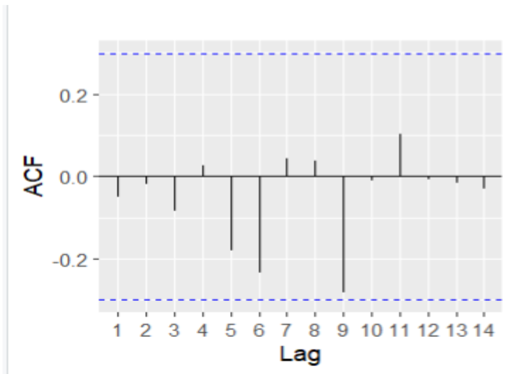
Final Model Chosen: ARIMA(2,1,2)

After further experimentation with alternative configurations, the ARIMA(2,1,2) model was identified as the best fit. This model successfully passed the Ljung-Box test, confirming the absence of significant autocorrelation in the residuals. The residuals appeared randomly distributed and remained within the confidence bands, indicating model adequacy. Additionally, the AIC value seems to be low (i.e., 206.59) compared to the other tested models, supporting its selection.

Thus, the ARIMA(2,1,2) model was chosen as the final model for India's inflation time series, based on its strong diagnostic performance and improved goodness-of-fit.

```
> fit2 <- Arima(India_ts, order = c(2,1,2))
> fit2
Series: India_ts
ARIMA(2,1,2)

Coefficients:
         ar1      ar2      ma1     ma2
      0.8910  -0.4571  -1.4470  1.0000
s.e.  0.1504   0.1504   0.1083  0.1359

sigma^2 = 6.282:  log likelihood = -98.3
AIC=206.59   AICc=208.26   BIC=215.28
```

```
> checkresiduals(fit2)

        Ljung-Box test

data:  Residuals from ARIMA(2,1,2)
Q* = 9.7615, df = 5, p-value = 0.08228

Model df: 4.   Total lags used: 9

>
```



2. China – Inflation Time Series (china_ts)

Basic Modeling using auto.arima()

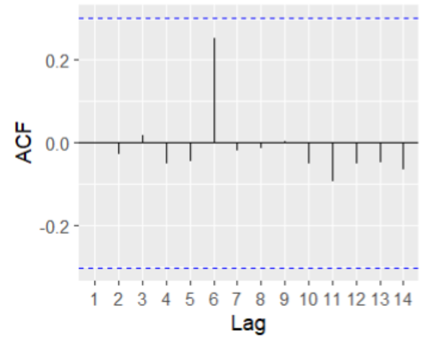
The inflation dataset was filtered to obtain the China-specific entries, which were then converted into a time series object. The auto.arima() function was applied to determine the optimal ARIMA model based on information criteria and diagnostic statistics.

Model Selection and Diagnostics

The auto.arima() function recommended an ARIMA(1,1,2) model for China's inflation series. The model showed good statistical properties. The results of the Ljung-Box test confirmed that the residuals were not significantly autocorrelated and implied that the model had captured the underlying dynamics of the series sufficiently. Residual diagnostics also showed that the residuals acted like white noise and were well within the confidence intervals. Secondly, the model had a low AIC score

Based on these diagnostic results, the ARIMA(1,1,2) model was selected as the final model for China's inflation series.

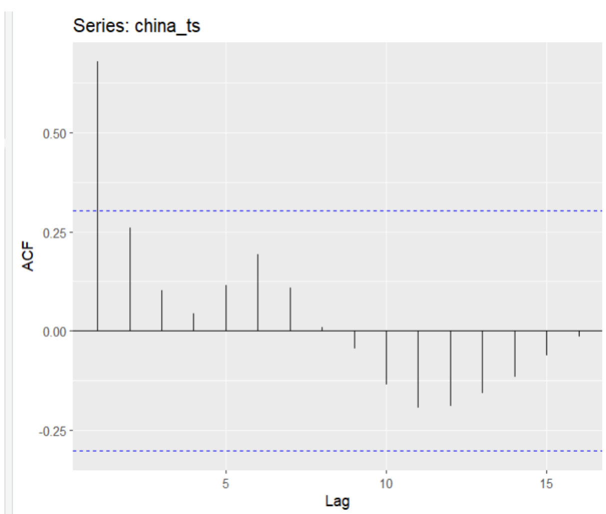
```
> fit3 <- Arima(china_ts, order = c(1,1,2))
> fit3
Series: china_ts
ARIMA(1,1,2)

Coefficients:
         ar1      ma1      ma2
      0.3540  -0.1737  -0.7162
s.e.  0.1999   0.1698   0.1253

sigma^2 = 14.3:  log likelihood = -112.01
AIC=232.02   AICc=233.13   BIC=238.87
```

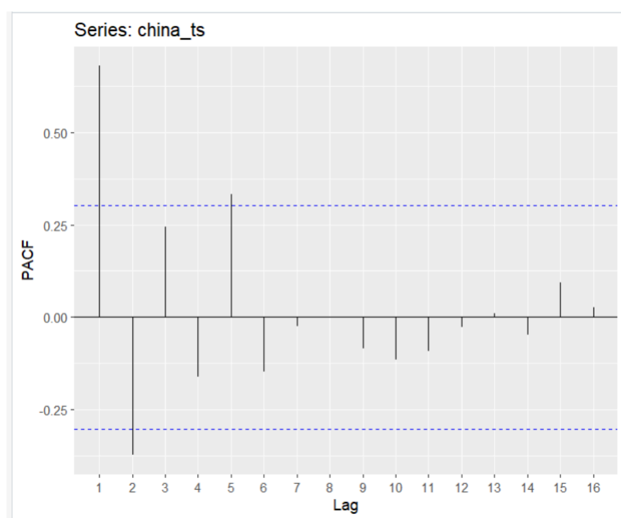



```
> checkresiduals(fit3)

        Ljung-Box test

data:  Residuals from ARIMA(1,1,2)
Q* = 3.5635, df = 5, p-value = 0.6138

Model df: 3.   Total lags used: 8
```

Here, P = 0.6138 > (0.05) So, here we fail to reject the null hypothesis of Ljung box and conclude that the autocorrelation did not exist between the residuals which is appropriate for ARIMA modeling.

The following are the ACF and PACF graphs for China:





3. United States – Inflation Time Series (US_ts)

Basic Modeling using auto.arima()

The time series for the United States was created by isolating the relevant inflation data and transforming it into a time series format. The auto.arima() function was applied to determine the most appropriate ARIMA model.

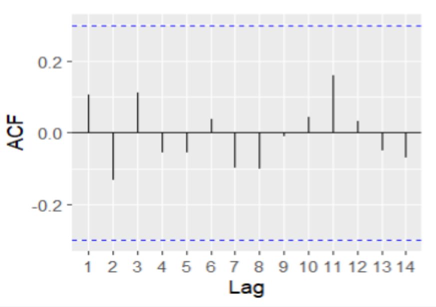
Model Selection and Diagnostics

The auto.arima() procedure suggested an ARIMA(0,1,0) model for the US inflation series, which corresponds to a random walk model. This model was found to be statistically adequate. The Ljung-Box test confirmed that there was no significant autocorrelation in the residuals. Furthermore, residual diagnostics indicated that the residuals were randomly distributed and fell within the expected confidence intervals. The model also produced a low AIC value, indicating a good fit.

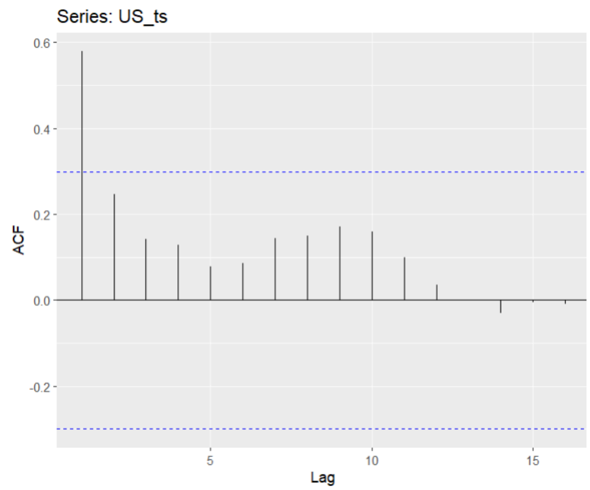
Given its simplicity and satisfactory diagnostic performance, the ARIMA(0,1,0) model was selected as the final model for the United States.

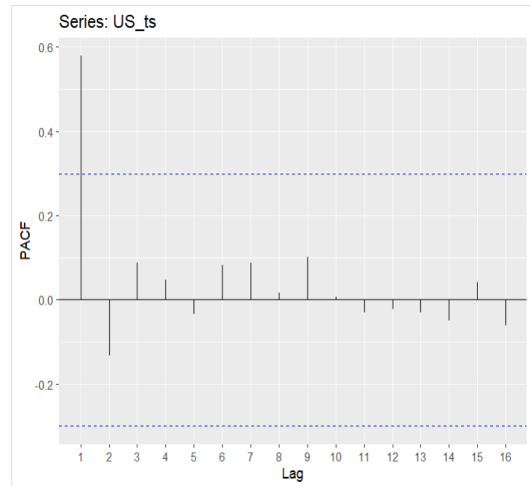
```
> fit4 <- Arima(US_ts, order = c(0,1,0))
> fit4
Series: US_ts
ARIMA(0,1,0)

sigma^2 = 2.021:  log likelihood = -74.37
AIC=150.74   AICc=150.84   BIC=152.48
>
```

```
> checkresiduals(fit4)

        Ljung-Box test

data:  Residuals from ARIMA(0,1,0)
Q* = 3.3917, df = 9, p-value = 0.9467

Model df: 0.   Total lags used: 9

>
```



The following are the ACF and PACF graphs for US:





Checking the Heteroskedasticity to find whether to apply GARCH or not?

We performed the ARCH LM test on model residuals, which indicated constant variance ($p > 0.05$), hence a GARCH model was not appropriate for this data.

In the ARCH LM test, hypothesis testing is used to determine whether heteroskedasticity (i.e., changing variance) is present in a time series and can be loaded with FinTs library. This helps decide if an GARCH-type model is appropriate for modeling the data.

The null and alternative hypotheses for the ARCH effect test are:

- **Null Hypothesis (H_0):** There is no ARCH effect — meaning the residuals exhibit constant variance (homoskedasticity). In other words, past squared errors do not help predict current variance.
- **Alternative Hypothesis (Ha):** There is an ARCH effect — the residuals show autocorrelation in squared terms, indicating that the variance is time-dependent (heteroskedasticity).

Checking ARCH LM test for India_ts:

```
> ArchTest(residuals(fit2))

	ARCH LM-test; Null hypothesis: no ARCH effects

data:  residuals(fit2)
Chi-squared = 13.906, df = 12, p-value = 0.3068

>
```

P>0.05 which implies there is no heteroskedasticity in India_ts data

Checking ARCH LM test for china_ts:

```
> ArchTest(residuals(fit3))

	ARCH LM-test; Null hypothesis: no ARCH effects

data:  residuals(fit3)
Chi-squared = 18.782, df = 12, p-value = 0.09393

>
```

P >0.05 which implies that there is no heteroskedasticity in china_ts data

Checking ARCH LM for US_ts:

```
> ArchTest(residuals(fit4))

	ARCH LM-test; Null hypothesis: no ARCH effect

data:  residuals(fit4)
Chi-squared = 2.7441, df = 12, p-value = 0.9971

>
```

P >0.05 which implies that there is no heteroskedasticity in US_ts data

As all the models for all data , suggesting that there is no ARCH effect. So, we are not proceeding with the GARCH model to fit the data

Results

1. Forecasting Inflation for India (2023–2025):

After fitting the model with

```
fit2 <- Arima(India_ts, order = c(2,1,2))
fit2
```

After evaluating the residuals of the selected model (fit2) and confirming that all necessary diagnostic conditions (discussed in the modeling section above)were satisfied, the model was used to generate forecasts for the next three years.

```
> forecast_fit2<-(forecast(fit2, h=3))
> forecast_fit2
     Point Forecast    Lo 80    Hi 80      Lo 95    Hi 95
2023       6.124615 2.844987 9.404243  1.1088564 11.14037
2024       6.057961 2.431245 9.684677  0.5113777 11.60454
2025       5.484506 1.496769 9.472242 -0.6142109 11.58322
```

Fig: Forecasting the expected inflation for India

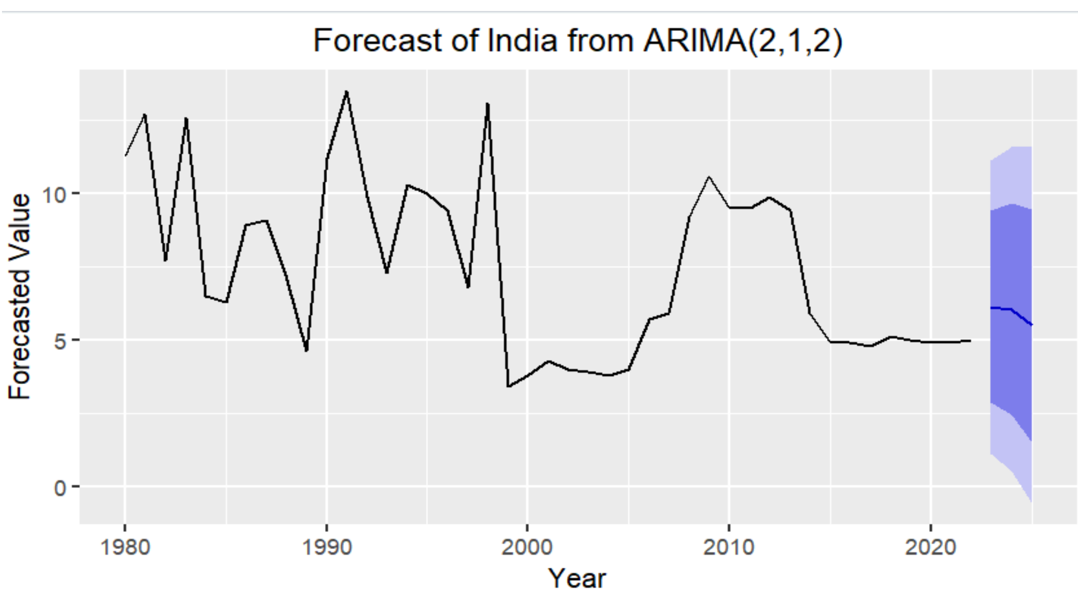


Fig: Forecast graph for India

The above figure presents the inflation forecast for India using the ARIMA(2,1,2) model. The black line shows the historical inflation trend from the time series object India_ts, and the shaded area represents the projected forecast values for the years 2023 to 2025. The bands reflect the 80% and 95% confidence intervals, which help quantify uncertainty in the forecasts.

The forecasted point values are:

- 2023: 6.124615
- 2024: 6.057961
- 2025: 5.484506

These values suggest a slightly declining inflation trend over the forecast horizon. The confidence intervals also narrow gradually across the three years:

- The 95% interval for 2023 is 1.1 to 11.14
- For 2025, it narrows slightly to -0.6 to 11.58

Accuracy of ARIMA(2,1,2):

```
> accuracy(fit2)
                     ME     RMSE      MAE       MPE     MAPE      MASE        ACF1
Training set -0.1447591 2.356102 1.835882 -7.250969 28.88468 0.9898208 -0.05131101
```

Interpretation:

- The ARIMA(2,1,2) model captures both autoregressive and moving average components, making it suitable for India's historically volatile inflation pattern.
- The forecast indicates moderation in inflation, with values stabilizing between 5%–6%.
- Despite the central forecasts being positive and steady, the wide confidence intervals reflect significant uncertainty, which is not uncommon in emerging economies like India that are sensitive to policy changes, commodity price shocks, and geopolitical factors.

Overall, this forecast provides a useful projection of India’s inflation direction while acknowledging the possible deviations due to external volatility. This highlights the need for cautious fiscal and monetary planning.

Forecasting Inflation for China (2023–2025):

After fitting the model with:

```
fit3 <- Arima(china_ts, order = c(1,1,2))
fit3
```

After evaluating the residuals of the selected model (fit3) and confirming that all necessary diagnostic conditions (discussed in the modeling section above)were satisfied, the model was used to generate forecasts for the next three years.

```
> forecast_fit3
     Point Forecast     Lo 80     Hi 80     Lo 95    Hi 95
2023       3.273347 -1.573715  8.120408 -4.139594 10.68629
2024       3.449281 -4.050268 10.948830 -8.020289 14.91885
2025       3.511565 -4.413408 11.436539 -8.608636 15.63177
```

Fig: Forecasting the expected inflation for China

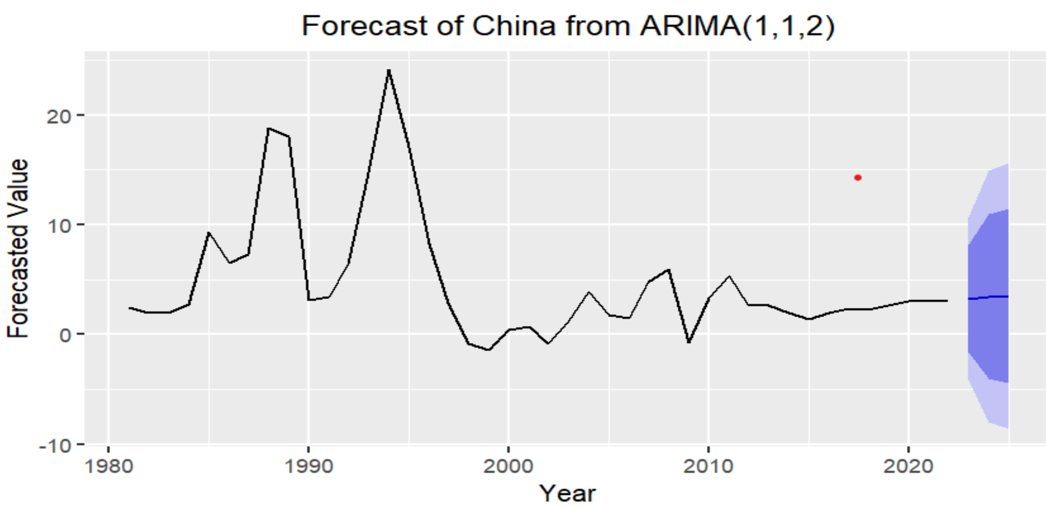


Fig: Forecast graph for China

The figure above displays the inflation forecast for China based on an ARIMA(1,1,2) model. The historical inflation time series (china_ts) is represented in black, while the forecasted values for the years 2023 to 2025 are shown along with shaded prediction intervals. The shaded bands represent the 80% and 95% confidence intervals, capturing the uncertainty around the future estimates.

The forecasted point values are:

- **2023**: 3.273347
- **2024**: 3.449281
- **2025**: 3.511565

Each point forecast is accompanied by an 80% confidence interval (e.g., for 2023: -1.57 to 8.12) and a 95% confidence interval (e.g., for 2023: -4.14 to 10.69). These intervals indicate the range within which the actual inflation values are expected to fall with respective levels of confidence.

Accuracy of ARIMA(1,1,2):

```
> accuracy(fit3)
                    ME     RMSE      MAE      MPE     MAPE      MASE         ACF1
Training set -0.225947 3.596816 2.462445 17.78481 104.6003 0.8505498 0.0009044635
> 
```

Interpretation:

- The model predicts a moderate and stable upward trend in inflation over the next three years, suggesting gradual economic recovery or increased demand-side pressure.
- However, the relatively wide confidence intervals reflect underlying uncertainty, possibly due to past volatility or external macroeconomic factors like trade disruptions or geopolitical tensions.
- Importantly, the lower bounds of the intervals are negative, indicating that deflationary conditions cannot be entirely ruled out in a worst-case scenario.
- This highlights the importance of continuously monitoring inflation indicators and refining models as new data becomes available.

Overall, the ARIMA model provides a data-driven outlook on inflation trends, and the confidence bands offer a realistic view of the uncertainty inherent in macroeconomic forecasting.

Forecasting Inflation for the USA (2023–2025)

After fitting the model with

```
fit4 <- Arima(US_ts, order = c(0,1,0))
```

```
fit4
```

After evaluating the residuals of the selected model (fit3) and confirming that all necessary diagnostic conditions (discussed in the modeling section above)were satisfied, the model was used to generate forecasts for the next three years.

```
> forecast_fit4 <- forecast(fit4, h = 3)
> forecast_fit4
     Point Forecast       Lo 80    Hi 80      Lo 95    Hi 95
2023            2.3  0.4781417 4.121858 -0.4862918 5.086292
2024            2.3 -0.2764967 4.876497 -1.6404117 6.240412
2025            2.3 -0.8555511 5.455551 -2.5259990 7.125999
>
```

Fig: Forecasting the expected inflation for USA

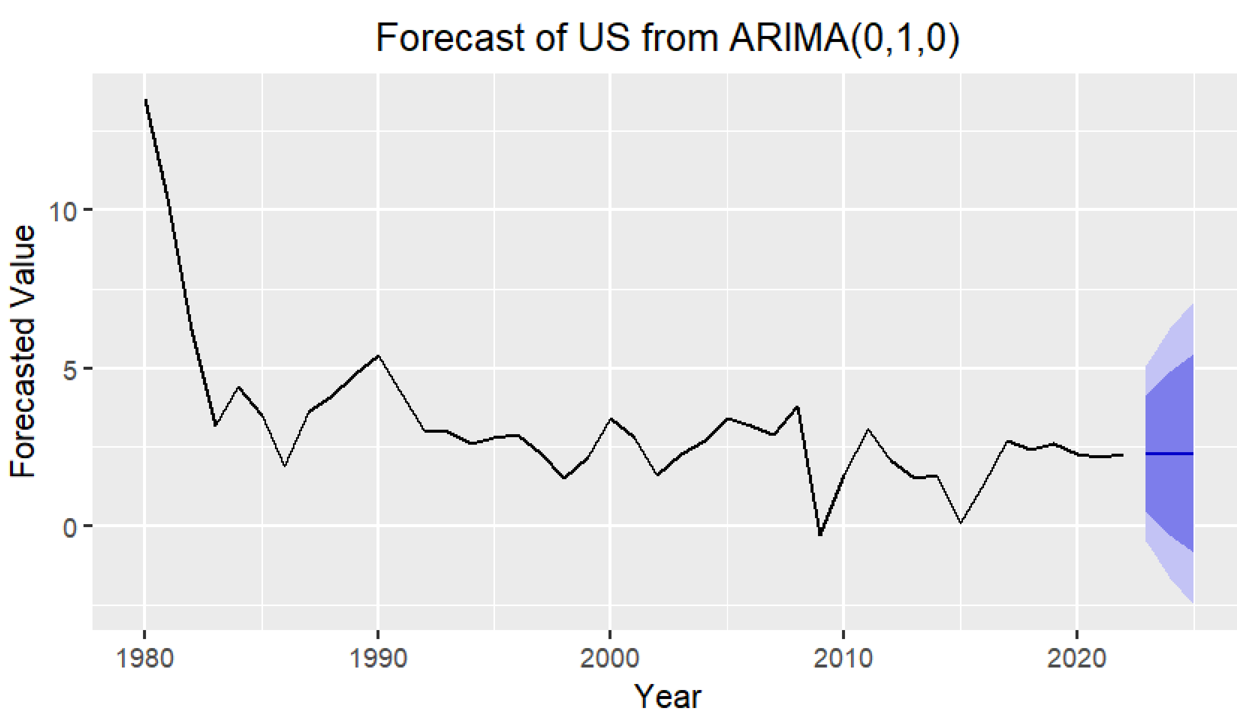


Fig: Forecast graph for USA

The figure above illustrates the inflation forecast for the United States using an ARIMA(0,1,0) model. The black line represents the historical inflation data (US_ts), and the shaded forecast area corresponds to the model's projections for the years 2023 to 2025. The shaded regions indicate the 80% and 95% prediction intervals, reflecting the range of possible future values with different levels of confidence.

The point forecasts are constant at:

- **2023**: 2.30
- **2024**: 2.30
- **2025**: 2.30

These values suggest a stable inflation expectation over the short term. However, the accompanying prediction intervals show increasing uncertainty:

- For 2023, the 95% interval ranges from -0.49 to 5.09
- By 2025, this range widens to -2.53 to 7.13

Accuracy of ARIMA(0,1,0):

```
> accuracy(fit4)
                     ME     RMSE      MAE       MPE     MAPE      MASE      ACF1
Training set -0.2601512 1.404976 1.004965 -5.101598 97.00687 0.9770494 0.1046818
>
```

Interpretation:

- The flat forecast line indicates that the ARIMA(0,1,0) model (a random walk with no drift) assumes inflation will stay close to the last observed value.
- The widening prediction intervals across the forecast horizon highlight that while the central estimate is stable, actual inflation outcomes may vary significantly due to unpredictable shocks or policy changes.
- The presence of negative lower bounds suggests a slim possibility of deflation, although this is less likely for a stable economy like the US.

Overall, the model provides a conservative forecast of inflation stability in the near future, but acknowledges the uncertainty inherent in macroeconomic projections. This reinforces the importance of continuous monitoring and adaptive policymaking.

Checking with General Forecasting Techniques:

To evaluate the performance we applied some benchmark models to our time series data using a three year forecast horizon.

```
naive_model <- naive(India_ts, h = 3)
autoplot(naive_model)

drift_model <- rwf(India_ts, drift = TRUE, h = 3)
autoplot(drift_model)
```

The drift model worked best overall with the lowest RMSE, negligible bias(ME~0) and the lowest MPE making it the most accurate of the two.

```
> accuracy(naive_model)
                ME     RMSE      MAE       MPE     MAPE MASE       ACF1
Training set -0.15 2.908485 1.854762 -9.745984 26.69245    1 -0.3853377
> accuracy(drift_model) #Best method on basis of accuracy result
                       ME     RMSE      MAE       MPE     MAPE     MASE       ACF1
Training set 2.85484e-16 2.904615 1.866667 -7.362834 26.73691 1.006418 -0.3853377
>
```

To compare the performance of our ARIMA model, we compared it with the best-performing model among our benchmarks, the Drift model. Based on training data, we compared the two models using major accuracy measures such as RMSE, MAE, MAPE, MASE, and ACF1

Now comparing with ARIMA:

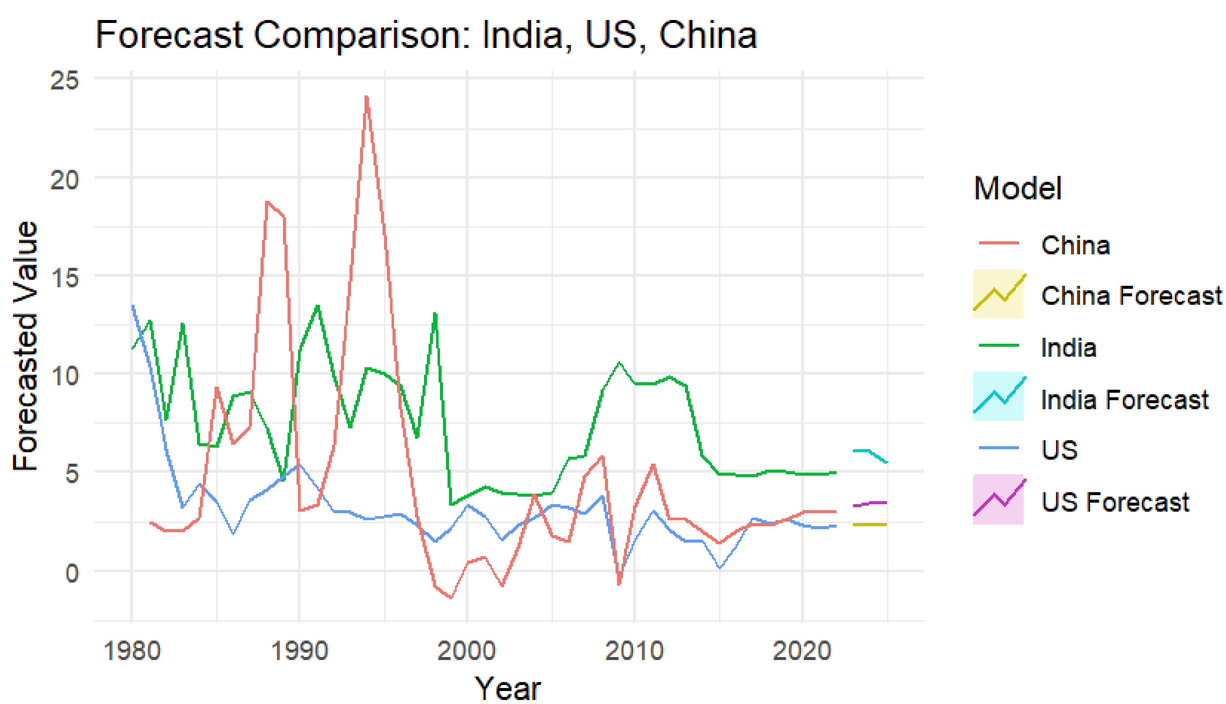
```
> accuracy_comparison <- bind_rows(accuracy_drift, accuracy_arima) |> select(Method, RMSE, MAE, MAPE,MASE,ACF1)
> print(accuracy_comparison)
# A tibble: 2 × 6
  Method  RMSE   MAE  MAPE  MASE    ACF1
  <chr>  <dbl> <dbl> <dbl> <dbl>   <dbl>
1 Drift   2.90  1.87  26.7 1.01  -0.385
2 ARIMA   2.36  1.84  28.9 0.990 -0.0513
```

The ARIMA model outperformed the Drift model on most of the metrics, especially in RMSE and MAE, showing more accurate predictions. Yet, the Drift model showed slightly inferior MAPE on some runs, which means although simple, it also produces competitive baseline forecasts.

Similarly we performed this for the US and China and we got similar results. ARIMA is better fitting as it has a lower RMSE compared to the general forecasting techniques.

Insights

The following graph gives a visual contrast of past inflation trends since 1980 and the projected inflation figures for India, China, and the United States. It points out crucial variations in the trend, volatility, and projected stability of inflation among the three big economies.



1. India's Continued Volatility and Gentle Upward Trend:
India's past inflation (green line) illustrates recurring and significant volatility from the 1980s to the early 2000s, consistent with economic liberalization shocks, supply-side bottlenecks, and policy changes. Although India's inflation stabilizes after 2010, the projection (light blue line) reveals a gentle upward trend, suggesting that future inflationary pressures continue to be challenging.
Implication: India, while getting better in managing inflation, is still exposed to domestic shocks like food inflation, oil price shocks, and foreign geo-political developments.

2. China's Early Volatility and Level Future Horizon:
China's inflation (red line) was highly volatile in the 1980s and 1990s, spiking sharply in the early phases of economic transformation from a centrally planned to a market economy. Conversely, the China forecast (yellow line) is near horizontal, implying tighter monetary management, stable growth strategies, and lower inflationary uncertainty over the past few years.
Implication: China has effectively shifted into a low-inflation regime, indicating policy sophistication and a more predictable macroeconomic framework in the future.

3. US Inflation Stability and Predictability:
The US inflation path (blue line) is highly stable relative to India and China. Inflation floats in a very narrow band except for minor shocks (such as the early 1980s recession, the 2008 financial crisis). The US projection (purple line) follows this stable pattern, with very little variation in inflation in the coming periods.
Implication: The Federal Reserve's persistent and credible inflation targeting policies have produced a low-volatility inflation environment, making the US the most inflation-stable economy of the three.

4. Forecast Comparison Reveals Divergent Risk Profiles
India's forecast indicates modest upward movement, reflecting higher inflationary risks compared to China and the US.
China's and the US's forecasts are nearly flat, reflecting low expected inflation volatility.
Key Insight: Emerging nations such as India might need more aggressive monetary and fiscal control than advanced economies to gain comparable inflation stability.

5. Structural Changes in Inflation Management
The transition from unstable to stable inflation in India and China underscores the long-run gains of macroeconomic reforms, central banking independence, and fiscal restraint.
US experience indicates that early and consistent measures to contain inflation build long-run certainty, shielding the economy even in global shocks.

Conclusions

This project explored inflation trends across three influential economies—India, China, and the United States—using time series analysis and ARIMA modeling. Through careful preprocessing of data from 1980 to 2022, the study aimed to not only understand past inflation behavior but also generate short-term forecasts that could offer insight into future economic trends. The use of ARIMA models was particularly suitable, as it allowed for capturing the dynamics of inflation data after ensuring stationarity through differencing and diagnostic testing.

In the case of India, the time series exhibited non-stationarity, which was addressed using first-order differencing. The ARIMA(2,1,2) model ultimately provided the best fit based on diagnostic results, and residual analysis confirmed no significant autocorrelation or heteroskedasticity, as supported by the ARCH-LM test. Similarly, China's inflation data was modeled using ARIMA(1,1,2), and the United States was best represented using an ARIMA(0,1,0) model. Across all models, residual checks and ARCH tests validated the robustness of the models by confirming white noise behavior and the absence of time-varying volatility, which made the use of GARCH models unnecessary.

Short-term forecasting for each country revealed distinct inflation trajectories, emphasizing the differences in macroeconomic behavior between developed and emerging economies. The combined plot comparing forecasts of all three countries provided a compelling visual comparison, showcasing the relative stability or volatility expected in the near future.

Overall, the project highlights the practical utility of ARIMA modeling in macroeconomic analysis, particularly for policymakers and financial analysts monitoring inflation trends. The step-by-step analytical approach—from stationarity testing to model validation and forecasting—demonstrates how time series forecasting can be both rigorous and insightful when supported by proper statistical testing and domain knowledge. This comparative study not only underlines India's evolving economic position but also provides a valuable benchmark against global inflationary patterns.

Critique

Looking at the process of undertaking this time series analysis project, some of the main insights, lessons, and limitations were apparent. This critique captures what was successful and the areas that could be improved upon in future versions.

What Went Well
One of the strengths of our project was the extensive methodology used. Right from data cleaning and data transformation to performing numerous stationarity tests (ADF, KPSS, ndiffs), we ensured that the series were properly ready for ARIMA modeling. The application of multiple diagnostics enabled us to cross-verify our results and arrive at solid conclusions regarding differencing and model selection.

Another of the strengths was the comparative perspective employed. Examining three vastly different economies—India (emerging), China (transitional), and the US (developed) - allowed us to look at how inflation behavior varies by economic context. This added depth to the analysis by providing sophisticated insights into economic actions under disparate structural conditions.

We also valued the interpretability and simplicity of ARIMA models. ARIMA models are widely acclaimed for their efficiency in forecasting in economics and enabled us to meaningfully interact with the fundamental ideas of autocorrelation, lag structure, and residual diagnostics.

Challenges Faced

In spite of the positives, we did face a number of difficulties. The preprocessing of data and transformation of the dataset from wide to long form were very time-consuming. Because for doing that, first we needed to understand the data properly and choose which columns to put .

The other significant challenge was model selection and validation. The ACF/PACF plots often failed to provide an obvious guideline for ARIMA orders. And several iterations we have tried to arrive at the best models utilizing AIC and diagnostic plots for comparison to support our choices more effectively.

In addition, the economic interpretation of the ARIMA components presented some challenges. Although we were able to statistically describe AR and MA terms, attributing them to actual drivers of inflation - like policy changes, oil prices, or geopolitical tensions - proved more challenging and would necessitate domain expertise.

What We'd Do Differently

If we were to repeat this project, we would put greater focus on out-of-sample testing and cross-validation. Dividing the time series data into training and test sets would give us a more objective platform for model performance evaluation.

We would also think about adding multivariate methods such as VAR, particularly to see how inflation in one nation can affect another—something highly salient in the globalized economy of today.

Visualizations could also be enhanced with interactive dashboards or dynamic plots, which would then render the trends and projections more interactive and informative for readers.

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